

M.TECH.
(SEM I) THEORY EXAMINATION 2022-23
ANTENNA THEORY AND DESIGN

Time: 3 Hours**Total Marks: 70****Note:** Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief. 2 x 7 = 14

- (a) What is the radiation resistance of a half wave dipole? Derive it.
- (b) What is the condition for an antenna to be frequency independent? Narrate.
- (c) What are the ways to increase the radiation resistance of a loop antenna.
- (d) Differentiate broad side and End fire array with mathematical support.
- (e) What is point source? Is it isotropic or non isotropic?
- (f) Why a Balun is used in an antenna?
- (g) Why a Cassegrain feed is used with a parabolic reflector?

SECTION B

2. Attempt any three of the following: 7 x 3 = 21

- (a) Define the followings
 - (i) Beam width
 - (ii) Front to Back Ratio
 - (iii) Beam Area
 - (iv) Directivity
 - (v) Effective Aperture
- (b) Prove that $l_{\text{eff}}(\text{transmitting})$ is same as $l_{\text{eff}}(\text{receiving})$ for any antenna where l_{eff} is effective length of an antenna.
- (c) With neat illustrations explain the geometry and requirements for a helical antenna radiating into Axial mode and give the relevant design relations.
- (d) Explain the array of N- sources of equal amplitude and spacing- Broad side case
 - (i) Direction of pattern maxima
 - (ii) Direction of pattern minima
 - (iii) Beam width of major lobe.
- (e) What is antenna synthesis? Explain the Schelkunoff method of synthesis for antenna array.

SECTION C

3. Attempt any one part of the following: 7 x 1 = 7

- (a) Describe micro strip antennas with neat diagrams. What are the different parameters effects the characteristics of micro strip antenna, explain.
- (b) What is Yagi-uda Antenna. Explain the construction and operation of Yagi-uda Antenna. Also explain its general characteristics.

4. Attempt any *one* part of the following:

7 x 1 = 7

- (a) What is an ideal dipole? Derive an expression for electric and magnetic fields of an ideal dipole.
- (b) Derive expression for Electric and Magnetic Field radiated by Hertzian Dipole and Sketch its Field Strength pattern.

5. Attempt any *one* part of the following:

7 x 1 = 7

- (a) With a suitable diagram explain the construction and principle of operation of a log –periodic antenna. Give the design equations and uses of log periodic antenna.
- (b) Obtain the expression for the field produced by TWA and compare its radiation with resonant antenna.

6. Attempt any *one* part of the following:

7 x 1 = 7

- (a) What is principle of pattern multiplication? Using the principal of pattern multiplication, estimate and sketch the pattern of a 8 isotropic elements of equal amp in phase and separated by $\lambda/2$ distance.
- (b) Explain the Woodward – Lawson synthesis for discrete linear array.

7. Attempt any *one* part of the following:

7 x 1 = 7

- (a) With a neat sketch explains the constructional features of parabolic reflector and obtain expression for its curved profile.
- (b) Explain calculation of antenna and scatterer characteristics in the method of moments.